

IMPROVING PATIENT APPOINTMENT ATTENDANCE AND HEALTHCARE SUPPORT USING DATA AND AI

Dataset Overview

The HealthConnect project focuses on improving patient appointment attendance and reducing missed appointments. HealthConnect is a fictional healthcare provider experiencing challenges including patients missing scheduled appointments, difficulty understanding factors associated with no-shows, inefficient use of appointment slots, and the need to improve patient engagement. The central project question is:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

For the Data Analytics track, the primary resource is the HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv). The dataset contains fictional and anonymized appointment records covering patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminder information, distance to the clinic, waiting time and appointment outcomes.

The Data Dictionary defines the dataset structure and identifies the following key variables:

- appointment_id – unique appointment identifier and primary key
- patient_id – anonymized patient identifier
- gender, age, age_group, appointment_type, booking_date, appointment_date, appointment_day, appointment_time, booking_lead_days, previous_appointments,
- previous_no_shows, reminder_sent, reminder_channel, distance_to_clinic_km, waiting_time_minutes, appointment_outcome

The main outcome variable is appointment_outcome, with the categories Attended, No-Show and Cancelled.

INITIAL DATA QUALITY ASSESSMENT

The initial quality assessment will focus on whether the data is complete, consistent and suitable for attendance analysis.

Unique records: Appointment_id is the primary key, so duplicate or missing appointment IDs should be checked before analysis.

Missing values: The Data Dictionary indicates limited missing values in distance_to_clinic_km and waiting_time_minutes. These should be identified and documented before analysis rather than automatically replaced.

Historical consistency: Previous_no_shows should not exceed previous_appointments. This business rule should be validated during data preparation.

Reminder consistency: Reminder_channel can be None when no reminder was sent. Therefore, this value should not automatically be treated as missing or erroneous.

Date consistency: Appointment Day is derived from appointment date, while booking_lead_days represents the number of days between booking and appointment. These fields should be cross-checked for consistency.

Outcome consistency: Attended, No-Show and Cancelled should remain separate categories. The Clinic Knowledge Base states that patients unable to attend should contact the clinic early to request cancellation or rescheduling.

IMPORTANT VARIABLES

The variables will be grouped according to their analytical purpose:

Category	Variables
Patient factors	Age, Age Group, Gender
Appointment factors	Appointment Type, Appointment Day, Appointment Time
Booking factors	Booking Date, Appointment Date, Booking Lead Days
Historical behaviour	Previous Appointments, Previous No-Shows
Reminder factors	Reminder Sent, Reminder Channel
Operational/accessibility	Distance to Clinic, Waiting Time
Outcome	Appointment Outcome

These variables will allow the analysis to investigate potential patterns associated with appointment attendance and no-shows.

BUSINESS QUESTIONS

The analysis will investigate the following questions:

1. What is the overall appointment no-show rate?
2. Which patient characteristics are associated with appointment no-shows?
3. Which appointment types, days and times have different attendance patterns?

4. Is previous no-show behaviour associated with future appointment attendance?
5. Is receiving an appointment reminder associated with attendance?
6. Do different reminder channels show different attendance patterns?
7. Is booking lead time associated with appointment no-shows?
8. Is distance to the clinic associated with appointment attendance?

These questions directly support the Data Analytics responsibility of investigating appointment attendance and no-show patterns.

PROPOSED KPIS

The Week 4 assignment requires **3–5 potential KPIs**, with each KPI linked to a business question. At this stage, the KPIs only need to be identified and justified; they do not need to be calculated or visualized.

Proposed KPI	Business Question	Relevance
No-Show Rate	What is the overall no-show rate?	Measures the scale of the missed-appointment problem.
Attendance Rate	How well are appointments being attended?	Measures successful appointment attendance.
Cancellation Rate	How frequently are appointments cancelled?	Separates cancellations from actual no-shows.
Reminder Coverage Rate	Is reminder coverage associated with attendance?	Measures the proportion of appointments receiving reminders.
Repeat No-Show Rate	Is previous no-show behaviour associated with future attendance?	Helps identify recurring missed-appointment behaviour.

INITIAL ANALYSIS APPROACH

The proposed approach is:

- 1. Data Understanding:** Review the dataset and Data Dictionary to understand the variables, categories and business rules.
- 2. Data Quality Assessment:** Check duplicates, missing values, invalid values, date consistency, historical appointment rules and reminder consistency.

3. Data Preparation: Standardize data types and categories and document any transformations. Original project resources will not be overwritten; cleaned or transformed data will be saved separately, as required by the project instructions.

4. Exploratory Analysis: Examine appointment outcomes across patient, appointment, booking, historical, reminder and operational variables.

5. KPI Analysis: Calculate and evaluate the five proposed KPIs in subsequent stages.

6. Insights and Recommendations: Identify important attendance/no-show patterns and provide evidence-based findings that can support future HealthConnect decisions.

Suitable tool for this track is Power BI.